NATIONAL UNIVERSITY OF TECHNOLOGY

(Department of Computer Science)

CS160– Database Systems

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Assignment 3

Total Marks: 10

Question [CLO: 2, PLO: 3]

The following tables form part of a Library database.

• Book (ISBN, title, edition, year)

• BookCopy (copyNo, ISBN, available)

• Borrower (borrowerNo, borrowerName, borrowerAddress)

• BookLoan (copyNo, dateOut, dateDue, borrowerNo)

Where,

• Book contains details of book titles in the library and the ISBN is the key.

• BookCopy contains details of the individual copies of books in the library and copyNo is the key. ISBN is a foreign key identifying the book title.

• Borrower contains details of library members who can borrow books and borrowerNo is the key.

• BookLoan contains details of the book copies that are borrowed by library members and copyNo/dateOut forms the key. borrowerNo is a foreign key identifying the borrower.

Formulate the following queries in relational algebra, tuple relational calculus, and domain

relational calculus.

1. List all book titles.

- **Relational Algebra:**
  π title(Book)
- **Tuple Relational Calculus:**
  { b.title | Book(b) }
- **Domain Relational Calculus:**
  { t | ∃ i, e, y (Book(i, t, e, y)) }

2. List all borrower details.

- **Relational Algebra :**
  Borrower
- **Tuple Relational Calculus:**
  { b | Borrower(b) }
- **Domain Relational Calculus :**
  { (n, nm, ad) | Borrower(n, nm, ad) }

3. List all book titles published in the year 2012.

- **Relational Algebra :**
  π_title(σ_year = 2012(Book))
- **Tuple Relational Calculus :**
  { b.title | Book(b) ∧ b.year = 2012 }
- **Domain Relational Calculus :**
  { t | ∃ i, e (Book(i, t, e, 2012)) }

4. List all copies of book titles that are available for borrowing.

Relational Algebra :
π_copyNo, title(σ_available = true(BookCopy ⋈ Book))

- **Tuple Relational Calculus :**
  { (bc.copyNo, b.title) | BookCopy(bc) ∧ Book(b) ∧ bc.ISBN = b.ISBN ∧ bc.available = true }
- **Domain Relational Calculus :**
  { (c, t) | ∃ i, e, y (BookCopy(c, i, true) ∧ Book(i, t, e, y)) }

5. **List all copies of the book title Lord of the Rings that are available for borrowing.**

   Relational Algebra (Times New Roman, 14pt):

   π_copyNo(σ_available = true ∧ title = 'Lord of the Rings'(BookCopy ⋈ Book))

- **Tuple Relational Calculus (Times New Roman, 14pt):**
  { bc.copyNo | BookCopy(bc) ∧ Book(b) ∧ bc.ISBN = b.ISBN ∧ b.title = 'Lord of the Rings' ∧ bc.available = true }
- **Domain Relational Calculus (Times New Roman, 14pt):**
  { c | ∃ i, e, y (BookCopy(c, i, true) ∧ Book(i, 'Lord of the Rings', e, y)) }

6. **List the names of borrowers who currently have the book title Lord of the Rings on loan.**

   Relational Algebra (Times New Roman, 14pt):

   π_borrowerName(BookLoan ⋈ BookCopy ⋈ σ_title = 'Lord of the Rings'(Book) ⋈ Borrower)

- **Tuple Relational Calculus (Times New Roman, 14pt):**
  { br.borrowerName | BookLoan(bl) ∧ BookCopy(bc) ∧ Book(b) ∧ Borrower(br) ∧ bl.copyNo = bc.copyNo ∧ bc.ISBN = b.ISBN ∧ bl.borrowerNo = br.borrowerNo ∧ b.title = 'Lord of the Rings' }
- **Domain Relational Calculus (Times New Roman, 14pt):**
  { n | ∃ c, i, d1, d2, a, e, y, bn (BookLoan(c, d1, d2, bn) ∧ BookCopy(c, i, a) ∧ Book(i, 'Lord of the Rings', e, y) ∧ Borrower(bn, n, _)) }

7. **List the names of borrowers with overdue books.**
   (Assuming today's date is '2025-04-08')

   Relational Algebra (Times New Roman, 14pt):

   π_borrowerName(σ_dateDue < '2025-04-08'(BookLoan) ⋈ Borrower)

- **Tuple Relational Calculus (Times New Roman, 14pt):**
  { br.borrowerName | BookLoan(bl) ∧ Borrower(br) ∧ bl.borrowerNo = br.borrowerNo ∧ bl.dateDue < '2025-04-08' }
- **Domain Relational Calculus (Times New Roman, 14pt):**
  { n | ∃ c, d1, d2, bn, a (BookLoan(c, d1, d2, bn) ∧ Borrower(bn, n, a) ∧ d2 < '2025-04-08') }